EXPERIMENT – 2

AIM :

To write a Python program to solve the 8-Queen Problem by placing eight queens on a chessboard so that no two queens attack each other.

ALGORITHM:

1. Start.
2. Initialize an empty 8 × 8 chessboard.
3. Place a queen in the first row.
4. Check whether the position is safe from other queens.
5. If safe, place the queen and move to the next row.
6. If no safe position is available, backtrack to the previous row.
7. Repeat until all eight queens are placed safely.
8. Display the chessboard.
9. Stop.

PROGRAM :

Experiment 2.py - D:/Abhi/Collage/AI/AI-Lab experiment-2026/Experiment 2.py (3.14.3)

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```
N = 8
board = [-1] * N

def safe(row, col):
    for r in range(row):
        if board[r] == col:
            return False
        if abs(board[r] - col) == abs(r - row):
            return False
    return True

def solve(row):
    if row == N:
        return True

    for col in range(N):
        if safe(row, col):
            board[row] = col

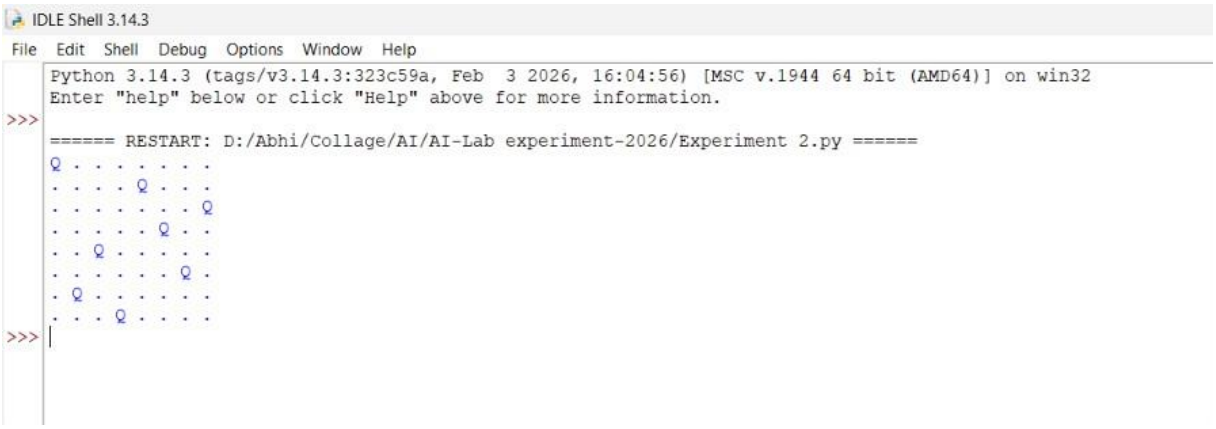
            if solve(row + 1):
                return True

            board[row] = -1

    return False

if solve(0):
    for r in range(N):
        for c in range(N):
            print("Q" if board[r] == c else ".", end=" ")
        print()
else:
    print("No solution")
```

OUTPUT:



RESULT:

Thus, the Python program to solve the 8-Queen Problem was executed successfully, and a valid arrangement of eight queens was obtained.